

What is variable?

A variable is a property that can take only values

Ex: Age = 14, Age = 25, Age = 100, here age is an variable which is holding different values.

Ages = [24, 25, 27, 30, 28]

Different types of variables:

There are 2 types of variables are there.

1. **Quantitative variable**
2. **Qualitative variable**

Quantitative variable:

Which we can measured numerically (Mathematical Operations)

Ex: Age, weight, height, rainfall (cms), temperature, distance etc.

Qualitative variables: Categorical variables

Based on some characteristics, they are grouped together

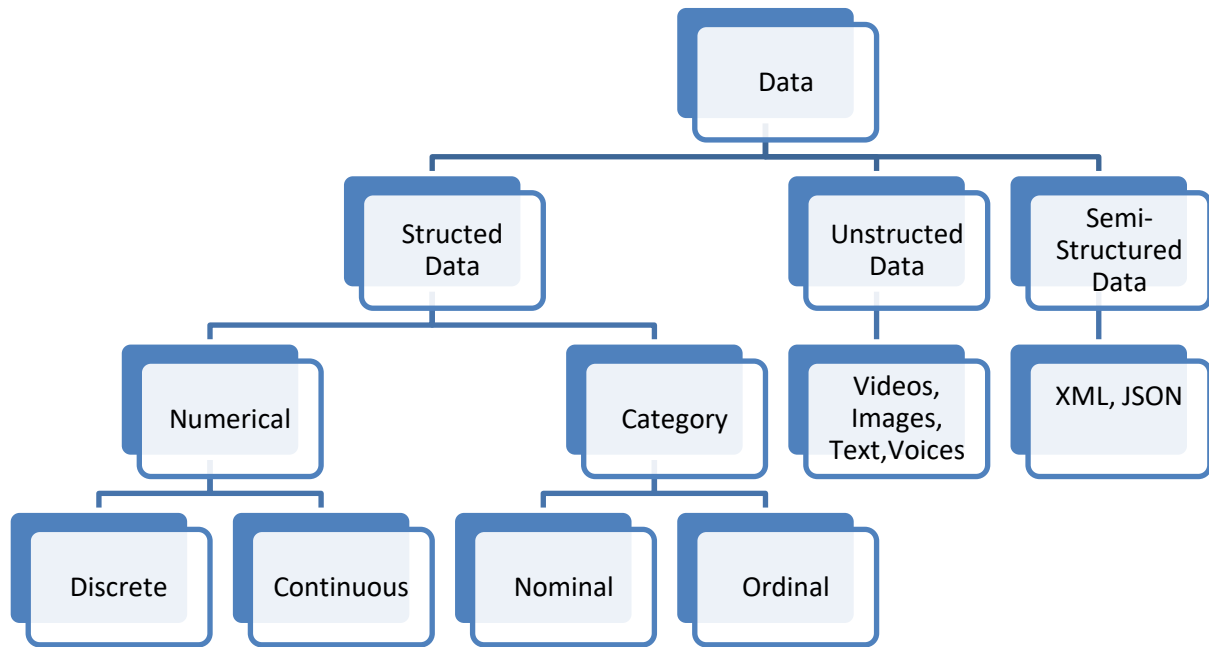
Ex: Gender, types of flower, types of movies etc.

Quantitative variable is further divided into **2 types**

- **Discrete variable - Whole numbers. Ex:** number of bank a/cs -1,2, number of children -1,2,3
- **Continuous variable – Decimal Numbers. Ex:** Height, weight, age, rainfall, speed

Exercise:

- What kind of variable is Marital status – **Categorical**
- What kind of variable is Gender - **Categorical**
- What kind of variable is Ganga river length – **Continuous**
- What kind of variable is Movie duration - **Continuous**
- What kind of variable is Pincode – **Discrete**



Let us discuss some more examples for better understanding of different variables.

Weight	Height	BMI
70	170	22
80	180	24
90	190	26
100	200	30

In the above example, we have **structured data** with all continuous variables (**weight, height and BMI** are **continuous variables**)

Lets discuss another example.

Name	Age	Height	Gender	Weight	Education
Sunny	25	170	Male	70	UG
Arijit	30	180	Male	80	PG
Prijam	35	160	Male	60	UG
Priya	20	150	Female	55	PhD
Aditi	27	145	Female	58	PG

Explanation:

- **Name** – Categorical – Nominal (Order is not matter)
- **Age** – Numerical – Continuous
- **Height** – Numerical – Continuous
- **Gender** – Categorical – Nominal
- **Weight** – Numerical – Continuous
- **Education** – Categorical – Ordinal (Where order plays a major role. Ex: UG, PG and PhD is the sequence)